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Module 1 Summary

After completing all of the materials, please take a moment to reflect on what was covered. The learning objectives below capture the key takeaways from this module:

- Define statistical learning and give reasons why one should study it.
- List and identify the three elements of a statistical learning problem.
- Explain why there can be a difference between points on a model and points in a dataset.
- List the 10 types of data as well as subtypes.
- For each type of data, produce examples.
- Describe how image data is represented in a computer.
- Define what a variable is as well as its two subtypes.
- List and describe the four types of statistical learning models.
- Explain the difference between supervised and unsupervised learning.
- Define what model evaluation is and explain why we need it.
- List the challenges of model selection and its connection to the bias-variance tradeoff.
- List the practical steps in statistical learning.
- Define the loss function.
- Define the expected prediction error, $EPE(f)$, and explain its components: the irreducible and reducible error
- Explain why we can optimize f by minimizing the $EPE(f)$ pointwise.
- Explain why f arises from a pre-defined family of functions rather than all possible functions.

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